

Project 1b

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ABSTRACT

WolfBite is an inherited Flutter/Firebase shopping assistant for WIC participants and caregivers. Project 1a established the system’s existing behavior and limitations through traceable use cases and independent testing. In Project 1b, we combined analyses from multiple LLMs with verified competitor sources, participant evidence, Project 1a engineering results, feasibility review, and relevant regulations, standards, and licenses to assess the market, identify product opportunities, and consider whether to continue with WolfBite or pivot. The evidence indicates a medium-confidence opportunity to help shoppers respond when an item that appears eligible is rejected at checkout. The team chose to continue developing WolfBite. Project 2 will pursue the finalized one-month milestones: clearer nutrition explanations, bounded basket-wide suggestions, a more usable shopping interface, and rule-based help for simulated checkout-rejection scenarios, including a baseline comparison of the checkout-help workflow. This plan extends the inherited product while keeping uncertain eligibility and checkout information explicit.

1 INTRODUCTION

Project 1a established the “Before” baseline for the inherited WolfBite application through code-traceable use cases and automated testing. Project 1b used that baseline to determine whether the team should extend WolfBite or pivot. To support this decision, the team compared four LLM analyses with verified competitor sources, participant evidence, feasibility findings, and relevant technical and regulatory material. This work produced a market survey, a documented product gap, a mission and stakeholder analysis, scoped engineering milestones, and an assessment of both continuation and pivot options.

The team decided to continue with WolfBite and selected the finalized “Now” milestones as the Project 2 plan. These milestones cover evaluation of the checkout-help concept, clearer nutrition explanations, basket-wide suggestions, interface improvements, and bounded help for simulated item rejections. The following sections present the evidence and constraints behind that decision and define how the proposed version will improve on the Project 1a system.

2 D1: MARKET SURVEY

2.1 Method

We ran the shared market-survey Prompt 1 with four LLM analysts: Codex, Terra, Gemini, and local Ollama, to generate rival candidates. We retained a rival only when at least two models named it or one model supplied a live, relevant URL. We then checked official product sources and independently audited the three strongest rivals. Each final rival independently satisfies the live-URL qualification rule through an official product source.

2.2 Verified Rivals

Table 1 summarizes the three verified rivals.

2.3 Selected Gap and Audience

Gap: Our feature audit found no documented integrated, item-level explanation-and-recovery workflow in WICShopper, myWIC Mosaic, or Bnft for cases where an app indicates that a product is eligible but checkout rejects it. Such a workflow could connect the rejected item, the participant’s current benefit or food category, and purchase evidence to likely causes and actionable next steps.

Audience: WIC caregivers and participants shopping for their households, especially those managing young children while an item they expected to be covered is rejected at checkout.

This need is supported by participant research reporting difficulty identifying WIC-eligible foods and stigma during shopping and checkout [1, 2]. WIC caregivers also rank balance checking and barcode scanning among their highest-priority app features [7]. Together, this evidence suggests that eligibility uncertainty during shopping remains consequential for the target audience.

We also evaluated two alternative gaps. We rejected nutrition comparison because participant evidence for that need was comparatively weak. We rejected broader receipt reconciliation because all three rivals already provide some form of purchase or transaction history. The narrower explanation-and-recovery gap therefore has stronger support from both the competitor audit and participant evidence.

2.4 Counterevidence and Scope

The conclusion has **medium confidence**. WICShopper already provides item-level purchase history in some agencies, myWIC tracks purchases, and Bnft provides transaction history. Public documentation may also omit program-specific features. In one month, the team can build and test a WolfBite prototype using

Table 1: D1 validated WIC rivals.

Rival	Evidence it is real	What it does	Relevant limitation
WICShopper	Official App Store listing and vendor purchase-history documentation [3, 4]	Shows benefits, scans eligibility, provides food guidance, and records item/category purchase history in supported agencies	No documented workflow connects a scan/checkout mismatch to its likely cause and recovery steps.
myWIC Mosaic	Official App Store listing [6]	Shows benefits, tracks purchases, scans products, and supports shopping, appointments, and certification	No documented item-level explanation and recovery workflow for an unexpectedly uncovered item.
Bnft	Official App Store listing [5]	Shows balances, scans benefit eligibility, manages cards, and provides transaction history	No documented item-level reconciliation or explanation for conflicting scan and checkout results.

fixed mismatch scenarios, rule-based likely causes, and recovery guidance. Without state benefit-system data, the prototype must not claim to know an authoritative point-of-sale rejection reason.

3 D2: MISSION, IMPACT, AND STAKEHOLDERS

3.1 Mission Statement

WIC participants and caregivers can face uncertainty when a product appears eligible in an app but is rejected at checkout, a setting in which participant research has documented difficulty identifying eligible foods and stigma during shopping and checkout [1, 2]. That conflict can add confusion and friction during a household shopping trip. We propose extending WolfBite with an item-level explanation-and-recovery workflow that connects the rejected item’s UPC and package details, the participant’s current benefit or food category, and available eligibility or purchase evidence. For a bounded set of mocked mismatch scenarios, the workflow will distinguish app evidence from facts that only a retailer or WIC agency can confirm, then show rule-based likely causes and actionable next steps. This guidance is intended to help participants and caregivers choose a next action after an unexpected rejection while keeping uncertain information visible.

3.2 Measurable Claim and M0 Evaluation

M0 measurable claim: During Project 2 M0, we will test whether the explanation-and-recovery workflow produces a task-success rate of at least 80% and at least 20 percentage points higher than the baseline workflow. A trial succeeds when a test user selects the prepared scenario’s reference next step within 60 seconds without facilitator help. The baseline is a control version of the current WolfBite flow that shows the pre-checkout eligibility result and a mocked checkout rejection but provides no explanation or recovery guidance. The new and baseline flows will use the same counterbalanced, synthetic mismatch scenarios, so the evaluation requires neither live WIC data nor retailer point-of-sale access. These values are project targets to be measured at M0, not achieved results.

3.3 Stakeholders

The four Prompt 5 outputs used different labels for related roles. To avoid inflating the list with synonyms, Table 2 includes a stakeholder group when there is a direct equivalent between LLMs.

Table 2: Overlapping Prompt 5 stakeholders and requirements consequences.

Stakeholder	Need or fear	Design response
WIC participants, caregivers, and household representatives	Misleading eligibility guidance, uncertainty after a rejection, or the wrong household state	Identify the active household context and show evidence limits, likely causes, and a short recovery action for the selected item.
Infants, children, and pregnant or postpartum benefit recipients	Eligibility or nutrition guidance could be mistaken for advice about what is healthy or appropriate	Keep coverage guidance separate from health claims and represent missing information as unknown rather than zero or safe.
WIC clinic staff, program administrators, and agencies	Extra support burden or a prototype mistaken for official WIC authority	Show data source and freshness, and route unresolved cases to the appropriate agency contact.
Cashiers, authorized retailers, and store managers	Checkout delay, disputes, or incorrect UPC and package information when the app and register disagree	Treat the register outcome as authoritative, provide concise recovery guidance, and make no payment or point-of-sale integration claim.
State WIC IT teams, EBT processors, and card-system vendors	Unsupported integrations, security risks, or simulated balances presented as official data	Use mock adapters and simulated data, document the integration boundary, and make no live balance or payment claim.
Approved-product-list stewards, manufacturers, and product-data providers	Stale, copied, misattributed, or misidentified UPC, package, and category records	Retain source/version provenance and test stale, duplicate, and conflicting records before presenting a likely cause.
Privacy, security, and legal reviewers	Exposure of household, benefit, receipt, location, or diagnostic data and liability from unsupported claims	Minimize stored data, exclude card credentials, use privacy-safe logs, and state the prototype's limits at decision points.
Program regulators and compliance personnel	Misrepresented eligibility, unnecessary data collection, or exclusion of protected users	Trace eligibility statements to their source and test explicit privacy, accessibility, and nondiscrimination requirements.
Participants with disabilities or limited English proficiency, and their advocates	Camera-only controls, color-only status, small targets, or unclear language could block the recovery task	Preserve manual entry, pair color and icons with text, use semantic labels and plain language, and test text scaling and assistive navigation.
Receipt OCR and other outside data-service providers	Sensitive uploads, unsupported formats, or use outside service terms	Keep OCR outside the core recovery path; if retained, require consent, validate inputs, minimize retention, and provide a mock service.
Application developers and maintainers	Unreproducible failures, sensitive logs, or one-month scope growth	Use a bounded rule set, deterministic fixtures, and privacy-safe diagnostics.
Project sponsors, funders, and procurement decision-makers	Misleading public-benefit claims, hidden integration costs, or a prototype that cannot be maintained	Define a narrow pilot, measurable outcomes, and explicit external dependencies rather than promising statewide deployment.

4 D3: ENGINEERING PLAN

4.1 Before: Project 1a

- **Set up and analyze the inherited WolfBite application.** The team cloned, ran, and inspected the existing Flutter/Firebase project so later decisions were based on implemented behavior rather than only the product description.
- **Map the documented behavior to the implementation.** The team finalized 20 code-traceable use cases and connected their primary, alternative, and exception paths to the relevant production code.
- **Create and execute automated use-case tests.** The team authored 86 tests spanning UC1–UC20. The recorded Project 1a run produced 82 passes and four retained failures that became evidence for the next version.

- **Document traceability, failures, and coverage.** The Project 1a report and supporting records preserve the test results, failure classifications, use-case traceability, and the 77.5% executable-line coverage baseline.

4.2 Now: One-Month Build and Test

Our one-month goal is to make WolfBite a more complete, usable shopping app. We will improve nutrition explanations, add an optional basket-wide suggestion mode, and make the main screens easier to use. A small checkout-help feature retains D2's explanation-and-recovery goal; the nutrition improvements support the shopping experience rather than establish a new verified market gap. Benefit information and checkout rejection scenarios will remain simulated.

- **M0: Check whether checkout help is useful.** Compare the current flow with the checkout-help feature using the same prepared rejection scenarios. Keep D2's target: at least 80% of trials choose the reference next step within 60 seconds without facilitator help, with success at least 20 percentage points above the baseline. Build a script that scores trial logs and runs in CI; check it with known passing and failing examples. Synthetic logs test the script, not real usability. Report pilot results separately if volunteers are available, and record a re-plan decision if measured results miss the target.
- **M1: Better nutrition information and food-choice explainers.** Extend the existing nutrition panel and badges with units, serving or reference amounts, and short explanations of the values behind a label or comparison. Show available sodium, sugar, fiber, protein, and fat information, including tradeoffs between options rather than a single unexplained "healthier" score. Reuse the nutrition utility and product records, preserve missing values as unknown, and retain the measurement basis in saved basket items. Check educational wording against cited nutrition sources during implementation. Unit and widget tests must verify displayed values and units, explanation rules, comparable serving bases, and missing-data cases; unknown values must never earn a favorable badge or comparison.
- **M2: Basket-wide suggestions for more balanced food choices.** Add an optional "Basket balance" mode alongside individual substitutions. Use basket quantities and M1's nutrition data to summarize food-group variety and rank possible swaps by their effect on the whole basket. Start with a small mock catalog with known package amounts and food-group mappings, and explicit demonstration targets for sodium, added sugar, and fiber. Prioritize only comparisons with known, compatible units and package amounts; flag incomplete totals and skip suggestions that depend on missing measurements. Rank candidates against the basket's largest target gap, rather than ranking each food in isolation; show before/after values and any tradeoffs for up to three suggestions. Reuse the APL category lookup, restrict swaps to the same benefit category and compatible package size, and recheck simulated allowances before applying one shopper-confirmed swap at a time. Recalculate after each change. Tests must show that the same item can receive different suggestions in different baskets, quantities affect results, and previews leave the basket unchanged. Also test confirmed swaps, balance consistency, missing measurement data, and no suitable candidate. This is a shopping aid with illustrative targets, not a personalized diet plan or a claim that a basket represents daily intake.
- **M3: A clearer, more complete shopping interface.** Improve the existing scan, basket, and benefits screens with consistent typography, spacing, product cards, and clearly labeled primary actions. Make nutrition explanations and the two suggestion modes easy to find; distinguish covered and shopper-paid items with text as well as color. Add useful loading, empty, error, and retry states. Widget tests and

a manual walkthrough must cover scan-to-basket navigation, opening an explainer, previewing a basket suggestion, and checking benefits. Verify phone and desktop layouts, enlarged text, and screen-reader labels without clipped content or inaccessible controls. Existing quantity and navigation tests must continue to pass.

- **M4: Add straightforward help for a rejected item.** Add an action on a basket item that opens a short explanation and next step for prepared package-size, category-balance, and stale-information scenarios. Use simple rules over mock item and benefit records, with an "unable to determine" response when evidence is missing. Test each explanation and the full item-to-help flow, including returning to the basket without changing its contents. Label the advice as possible causes, not an official checkout decision; use this feature in M0's comparison.

Build the M0 script and M1's measurement handling and define M4's fixtures and flow in week 1. In weeks 2–3, implement the bounded basket suggestions while completing M4 and targeted M3 work. Reserve week 4 for integration and the evaluation. M4 and M0 are the critical path; if they slip, defer M2 extensions and nonessential M3 polish. Keep basket suggestions to one-swap previews over the mock catalog; multi-item optimization and personalized meal planning are outside this month. Add tests alongside each change, run relevant Project 1a regressions in CI, and document how another team can run the mock demo and extend the catalog, explanation rules, basket targets, and checkout-help cases.

4.3 Future: After the One-Month Project

- **Permissioned, versioned state APL synchronization.** Replace static eligibility fixtures with a monitored adapter that records jurisdiction, version, refresh time, conflicts, and stale-state information.
- **Partner-backed read-only evidence pilot.** Evaluate recovery with authorized participant or transaction evidence without approving purchases, processing payments, or modifying benefits.
- **Privacy-reviewed on-device receipt recognition and reconciliation.** Use local recognition where practical, connect item and quantity evidence to basket and benefit information, and retain a manual fallback.
- **Spanish and broader accessibility validation.** Add professionally reviewed Spanish content and test disability, literacy, language, device, and network conditions.
- **Production hardening and operational release.** Add access-control tests, monitoring, backups, retention and deletion policies, security review, and deployment-specific compliance work after scope and partnerships are stable.

4.4 Support Material

Table 3 separates material that governs the WIC program from technical and accessibility guidance that constrains this prototype and from licenses that govern redistribution. The design consequences are requirements for the Project 2 explanation-and-recovery workflow, not claims that WolfBite is a WIC agency, retailer system, or EBT processor.

Table 3: Support material and the Project 2 design checks it creates.

Support material and source	What it says that matters here	Effect on WolfBite's design
Federal regulation: 7 CFR Part 246, especially § 246.10(b) and (e)(12). USDA Food and Nutrition Service (FNS); eCFR text and FNS regulatory summary.	The federal rule supplies minimum food and package criteria. It also permits a state agency to add criteria and requires the agency to identify acceptable brands and package sizes. FNS therefore cautions that a state need not authorize every food that meets the federal requirements.	An eligibility result must identify its jurisdiction, exact UPC/package, source, and source date. Meeting only federal criteria will be labeled as general guidance, never as a guarantee of purchase. The workflow will distinguish that evidence from state authorization and from the retailer's actual checkout result.
State domain guidance: North Carolina WIC Authorized Product List (APL). NC Department of Health and Human Services (NCDHHS); APL and UPC-submission page and WIC Vendor Manual.	NCDHHS maintains an APL by UPC; products can be added, changed, or removed, and vendor systems must receive APL updates. A product that appears to fit a federal category is not thereby present in North Carolina's current APL.	The bounded NC scenarios will use a dated APL fixture keyed by UPC and package details. A missing, changed, or stale APL record may be shown only as a likely cause. Recovery will link to the current NC list and agency help or suggest checking an authorized alternative, rather than declaring the item approved or diagnosing the register.
Technical/domain guidance: WIC EBT Technical Implementation Guide and Operating Rules. USDA FNS; official resource page.	These resources standardize WIC EBT online purchase messages and file handling for state agencies, authorized vendors, processors, and other EBT stakeholders. They describe the transaction ecosystem; they do not turn a shopping assistant into an EBT system.	Fixtures and explanations will keep agency, APL, benefit, and retailer evidence separate. Project 2 will use mock adapters and no card number, PIN, live purchase message, or balance mutation. It will say "likely cause" when only the state or retailer transaction system could supply an authoritative result.
Accessibility standard/design target: WCAG 2.2 Level AA. World Wide Web Consortium (W3C); WCAG 2.2 and W3C mobile guidance.	Relevant success criteria prohibit color as the only means of conveying information (1.4.1), set minimum contrast (1.4.3) and target-size/spacing rules (2.5.8), require identified errors and known corrections (3.3.1/3.3.3), and keep repeated help consistent (3.2.6). W3C explains how WCAG guidance can also inform native mobile applications.	Approval, rejection, and uncertainty will use text plus an icon, not color alone. The flow will use semantic labels, readable contrast, scalable text, adequately sized or spaced controls, a manual-UPC alternative to the camera, a specific error and next action, and consistently placed agency help. Tests will cover text scaling and assistive navigation.
Application license: MIT License. The inherited WolfBite Project3/LICENSE.md; the same license is visible in the upstream repository.	MIT permits use, modification, and redistribution, provided the copyright and permission notice remain in copies or substantial portions; it also disclaims warranties.	We will keep the inherited copyright and MIT text with redistributed WolfBite code and will not describe the prototype as warranted or officially approved.
Selected dependency licenses. The declared packages are in Project3/pubspec.yaml; official package records confirm provider (MIT), mobile_scanner (BSD 3-Clause), and firebase_core (BSD 3-Clause).	MIT requires its copyright and permission notice. BSD 3-Clause requires its copyright, conditions, and disclaimer in source or binary distributions as applicable and prohibits using contributor names for endorsement.	Releases will preserve applicable third-party notices. Before adding an APL, accessibility, or scanning package, the team will record the resolved version and license, check compatibility, and reject any dependency whose redistribution duties the project cannot meet.

5 D5: LLM ANALYSIS AND VERIFICATION

5.1 Models, Prompts, and Evidence

The four model/runner pairings were GPT-5.6 Sol through Codex desktop, collected by Abigail Close; GPT-5.6 Terra through Chat-GPT, collected by Supreme Constantine; Gemini 3.1 Pro (High) through Antigravity, collected by Aditya Mahajan; and Llama 3 8B through local Ollama, run by Satwi Shah. Each model completed the required market-survey Prompt 1 and the three-part red-team Prompt 10. Keeper prompts are preserved in Project1b_Work/prompts/keeper_prompts.md, and raw transcripts are stored by model under Project1b_Work/model_outputs/. The Codex and Gemini P01 records contain the exact keeper; Terra and local P01 preserve the similar earlier wording they received. The finalized Gemini and local P10 outputs are usable but less reproducible: Gemini's saved file omits the pasted prompt text, and the local transcript contains a prompt placeholder rather than the exact text.

Prompt 1 generated candidates; it was not accepted as market proof. WICShopper appeared in all four outputs after name normalization, myWIC Mosaic in three, and Bnft in two. Live-source and scope checks produced the strongest direct set reported in D1: WICShopper, myWIC Mosaic, and Bnft. This comparison also

exposed the value of multiple models: Terra emphasized state fragmentation, Gemini corroborated several direct products, Codex supplied the most complete live evidence, and the non-browsing local model produced mostly adjacent or unverified candidates that forced stricter checking.

5.2 Prompt-by-Model Comparison

All four models challenged market need, feasibility, and competitors, but they did not produce identical conclusions. Gemini called the plan indefensible unless M1–M3 were removed; Codex and Terra defended a bounded prototype with a cut order; Ollama did not recommend a pivot. The supported team conclusion is that P10 found no reason to abandon WolfBite. It did require an execution adjustment: M4 and M0 form the critical path, M1–M3 remain bounded supporting work, and secondary scope must yield if the core flow or evaluation slips. The main remaining risk is whether participants will use non-authoritative help at checkout; M0 tests that question rather than assuming the answer.

5.3 Wrong Outputs and How We Caught Them

The team checked candidate identity, URL relevance, product scope, and feature claims separately instead of accepting model prose as evidence. Concrete corrections included:

Table 4: Cross-model Prompt 10 comparison and team adjudication.

Model	Strongest useful attack	Evidence judgment	Team response and model limitation
Codex	Checkout friction does not prove demand for this exact workflow, and M1–M3 may crowd out the M4-to-M0 critical path.	Supported: direct demand and deployed-rival behavior remain unverified; the dependency/scope risk is real.	Treat WolfBite as an evaluation-first prototype and defer secondary scope if needed. Codex could not directly observe users or deployed rivals.
Terra	Agencies, processors, retailers, and POS systems own the authoritative evidence WolfBite lacks; the roadmap also exceeds the core mission.	Supported as a structural constraint, although those institutions are not direct software rivals.	Keep claims non-authoritative, use fixed scenarios, and prioritize M4/M0. Terra’s competitor category was broader than D1’s product definition.
Gemini	Five milestones are too much for roughly 160 hours, and WICShopper is better positioned through program-connected data.	The scope concern is supported; the absolute “impossible” verdict and demand to abandon M1–M3 were not backed by measured estimates.	Preserve the agreed milestones in bounded form with an explicit cut order protecting M4/M0.
Local Ollama	Established rivals have more development, testing, and adoption, so WolfBite needs a clear differentiator.	Rival maturity is real, but its claim that mismatch recovery is an edge case had no prevalence evidence; it missed the detailed scope conflict.	Retain the medium-confidence gap and test usefulness at M0. The output was generic and its saved prompt was incomplete.

- Gemini P01 returned seven products rather than ten, and its linked listings did not establish its scanner, login, or reinstall-frequency claims. We kept verified product identities but discarded those weaknesses.
- Terra P01 supplied source labels instead of URL strings. Its uncorroborated Indiana WIC, Wisconsin MyWIC, and SC WIC entries therefore failed the team’s evidence rule.
- Live checks rejected local P01’s FreshConnect, generic “WIC Mobile App,” StoreWise, and Grocery IQ entries and reclassified several rebate/list applications as adjacent. Product pages also failed to support its claimed receipt scanning for Out of Milk and AnyList.
- An initial team note incorrectly reported zero Terra/local overlap and treated the local WICShopper URL as fabricated. Name normalization and the live redirect showed the product was real, so WICShopper was credited to all four models.
- Gemini P10 called the full plan impossible without supplying an effort breakdown, while local P10 called mismatches an edge case without prevalence data. We retained both as risks to test, not verified findings; neither claim overrode the inherited code evidence or the team’s bounded scope decision.

These corrections prevented unsupported products and weaknesses from becoming requirements. They also narrowed the final D1 claim from broad nutrition or receipt novelty to a medium-confidence explanation-and-recovery gap.

5.4 Most and Least Useful Prompts

The most useful prompts were the custom focused-gap validation (P07) and the required red team (P10). P07 separated competitor absence from participant demand, rejected nutrition comparison and broad receipt reconciliation as the primary gaps, and produced the narrower medium-confidence gap. P10 independently exposed the demand, authority, and scope risks and led to the M4/M0 priority rule. P01 was essential but limited: it generated and corroborated rivals, yet every answer required source checking. The least useful for the final decision was P06’s three-futures exercise. Its SAFE/BOLD/WILD concepts supported ideation, but speculative

OCR and community directions supplied no market validation and did not determine the selected gap or build plan.

5.5 Model Strengths and Weaknesses

Across P01 and P10, the models contributed different strengths and failure modes:

- **Codex** had the strongest source completeness and evidence-boundary reasoning, but produced broad, verbose analyses still dependent on external user and competitor validation.
- **Terra** gave the strongest critique of roadmap alignment and authoritative data, but its P01 evidence fields lacked URLs and its P10 competitor category extended beyond competing software.
- **Gemini** gave the clearest, most forceful scope challenge, but its P01 was incomplete and its P10 feasibility verdict was overconfident.
- **Local Ollama** completed the attack structure locally and named the verified rivals in P10, but its non-browsing P01 was factually weak and its P10 was generic and incompletely preserved.

The disagreement improved the analysis without being manufactured: it exposed both factual verification failures and a real policy choice between removing M1–M3 and retaining them behind an M4/M0 scope gate.

6 CONCLUSION

Project 1b identified a medium-confidence opportunity to help WIC participants when an app marks an item as eligible but checkout rejects it. WICShopper, myWIC Mosaic, and Bnft cover much of the shopping process, but our review found no documented item-level explanation and recovery workflow.

Using four LLMs helped us find and compare more competitors, while our own checks removed claims we could not verify. Prompt 10 did not give us a reason to change projects, but it warned that our plan may be too large for one month and that WolfBite cannot access official checkout data. Other milestone and feasibility prompts showed that the project should still be manageable if we use mock scenarios, set clear goals, and focus first on M4 and

M0. In Project 2, we will follow this plan, cut lower-priority work if needed, keep our Project 1a tests working, and evaluate whether the checkout-help feature is actually useful.

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